Name
Date
Arithmetic

Test IV (on classes 25-28)

Instructions

I) Write a title, name, date, and course.
II) Number each problem, circle the number, rewrite the problem, and show all work.
III) Put a rectangle around your answers.
IV) Write any fractions in your answers in lowest terms and never leave your answers as improper fractions.

Average Number of Hours Per Day that Seven People Watch TV:

2, 2.1, 0.7, 4.1, 2.4, 2.1, 2

1 Find the mean, median, mode, and range of the TV data above.

2 Find the interquartile range of the TV data above and display the data with a dot plot and a box and whisker plot; then, identify any outliers. If there are no outliers, write "No Outliers."

3 Convert $5/9$ to a decimal and round to the nearest hundredth.

4 The ratio of pine trees to elm trees to cedar trees is 2:4:9. If there are 36 cedar trees, how many elm trees are there? How many pine trees are there?

Solve the following proportions:

5 $\frac{2}{5} = \frac{8}{n}$

6 $\frac{b}{4} = \frac{4}{5}$

7 Write a ratio using the numbers 5 and 8 in six different ways.

8 Estimate the sum below by rounding the first number to the nearest ten thousand and the second number to the nearest thousand; then add.

85,699 + 5,600

9 Is the proportion below true?

$\frac{36}{40} = \frac{27}{30}$

10 Round the following numbers to the nearest thousand.

a) 8,099.6 b) 399 c) 90,657.001 d) 512,000

11 Write the following ratios in lowest terms.

a) $\frac{4}{8}$ b) 10 : 14 c) 24 to 15 d) $\frac{49}{63}$

12 If Eddie drives 207 miles in 3 hours, how much time would it take him to travel 144.9 miles?

13 Round the following numbers to the nearest units place.

a) 832.6 b) 1,320.4444 c) 0.309 d) 9999.9999

14 Estimate the quotient below by first rounding both numbers to the nearest tenth and then dividing.

$1.55019 \div 0.6346$

(15) If the ratio of bananas to oranges is 3 to 2, and the sum of the number of bananas and the number of oranges is 40, how many bananas are there?

(16) Reduce the following rates to lowest terms.

a) 42 kilometers in 12 days. b) 15 arrows for every 6 bows.

(17) Convert the following rates to unit rates. Write your answers as decimals or fractions.

a) 60 flowers for every 10 square yards. b) 3 feet in 8 seconds.

(18) Round the following numbers to the nearest whole number.

a) 3,989.713 b) 2.803 c) 0.56 d) 453,899.111

(19) If the ratio of rhinos to hippos is 4 to 9, and there are 20 rhinos, how many hippos are there? Write a proportion statement to represent this problem and use it to solve the problem.

(20) Draw a line graph of the data below. Label the graph properly.

Viewership of the TV show Knights of the Square Table.

Week #	# of Viewers
1	40,000
2	56,000
3	72,000
4	120,000
5	450,000
6	800,000
7	750,000

(21) Draw a bar graph of the data below. Label the graph properly.

Number of Wins of Boxers

Boxer	# of Wins
Fabulous Finn	18
Knockout Nick	10
Perfect Prince	15
Loud Larry	16
Brutal Bennie	7
Donny Doherty	11
Bill Olsen	10
Bull Booker	9

(22) If it takes 21 hours to walk 10 miles, how long will it take to walk 6 miles? You can write your answer as a mixed number or as a decimal.

(23) Solve the following proportion.

$$\frac{18}{30} = \frac{c}{10}$$

Distance Six Triathletes Run per Day when Training (miles):

1.8, 2.5, 3.1, 2.7, 2.4, 2.6

(24) Find the mean, median, mode, and range of the triathlete data above.

(25) Find the interquartile range of the triathlete data above and display the data with a dot plot and a box and whisker plot; then, identify any outliers. If there are no outliers, write "No Outliers."